

Handheld Vacuum Pass and Dock SOP (1x Rug Zone, in situ)

One episode vacuums one small rug where it lies, then puts the vacuum away. The vacuum is unclipped from its dock, switched on, run over the rug in three overlapping lanes and one edge pass, switched off, emptied into the bin, and docked again until it clicks.

The base is **passive**. It has no drive of its own. It is pushed by hand up to the front of the platform and locked there. Nothing is carried away to a table. Everything the episode touches is already on the platform when recording starts.

The rug is worked **as found**. It lies flat on the platform with a scatter of dry crumbs over the whole of it. The three things the episode uses are the handheld vacuum, its dock, and the bin. They all sit on the platform around the rug. The episode ends with the rug clear of crumbs, the crumbs in the bin, and the vacuum clicked back into its dock with its power light off.

This is an in-situ task. Four rules follow from that.

First, the platform is a fixed piece of furniture. So **every approach is from the front, level**. The arms come in over the front edge of the platform and go back out the same way. Nothing swings down from above.

Second, **the platform, the rug, the dock, and the bin are never loaded**. No gripper, wrist, or forearm leans on them. The nozzle is pushed only hard enough to keep it flat on the rug. The rug is never dragged or bunched.

Third, **the nozzle stays on the rug while the vacuum runs the pattern**. From the end of lane 1 to the end of the edge pass it is slid, never lifted. It leaves the rug only once, to be carried back to the switch spot.

Fourth, **nothing is carried across the bin or the dock**. The vacuum travels between the dock, the rug, and the bin along the front of the platform, level and low.

The order never changes. Unclip the vacuum. Switch it on. Run three lanes. Run the edge pass. Switch it off. Empty the dust cup into the bin. Dock the vacuum until it clicks. The vacuum is switched off before it goes near the bin, with no exception.

The two arms never cross. The rug lies in front of the right arm. The **right gripper works the rug and the dock**. The **left gripper works only two places**: the power button while the nozzle rests at the switch spot, and the dust cup while it hangs over the bin. The left gripper always stays left of the right gripper.

Arm roles are fixed and never swap. The **right gripper is the tool gripper**. It lifts the vacuum out of the dock. It holds the handle for every lane, the edge pass, the switch-on, the switch-off, and the emptying. It clicks the vacuum back into the dock. The **left gripper is the button**

gripper. It presses the power button once to start and once to stop. It presses the release catch so the cup flap opens over the bin, and it pushes the flap shut. It never touches the handle, the nozzle, the dock, or the rug.

The lanes run **from the left edge of the rug rightward**, so each new lane is laid beside the one just done and the nozzle never travels back over rug it has already cleaned.

Setup

Complete the base positioning and both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Base positioning

The base is **passive**. It has no drive of its own. It is pushed into place by hand. It is never steered, nudged, or repositioned once recording starts. It is parked once, before recording, and does not move again until the episode is over.

1. Push the base by hand up to the front edge of the platform. Stop it **square to the front edge**, so the edge runs straight across the frame and neither end of the platform sits nearer than the other.
2. Stop it so the **left edge of the rug lines up with the middle of the base**. The rug then lies in front of the right arm, the bin in front of the left arm, and the dock out to the right.
3. Stop it **close enough** that the right gripper puts the nozzle flat on the rug at its back edge without the arm extending. Stop it **far enough** that no arm, wrist, forearm, or part of the base touches the platform edge, the bin, or the dock when both arms work at once.
4. Check the **rug reach**. With the base parked, the **right gripper** holds the vacuum by its handle and puts the nozzle flat on the rug at the front-left corner, the back-left corner, the back-right corner, and the front-right corner. It runs one full lane from the front edge to the back edge. It does all of this without extending.
5. Check the **dock reach**. With the base parked, the **right gripper** closes on the handle of the docked vacuum, lifts it straight up out of the dock, and lowers it straight back in until it clicks, without extending and without the wrist turning.
6. Check the **button reach**. With the base parked and the **right gripper** resting the nozzle at the switch spot, the **left gripper** reaches the power button from above and presses it, without extending and without reaching past the right gripper.
7. Check the **bin reach**. With the base parked, the **right gripper** holds the vacuum level over the bin with the dust cup above the middle of the bin's open top. The **left gripper** reaches

the release catch on the left side of the cup, and reaches under the open flap to push it shut. Both without extending.

8. Lock or brake the base. Push it firmly once by hand. It must not roll, creep, or turn.
9. If any of lines 1 to 7 fails, push the base to a new park by hand and start again at line 1. If line 4 still fails after a re-park, the rug is too wide or too deep to work from one park. Do not work it.

The base stays locked and still for the whole episode. Nothing moves it. No arm leans on the platform hard enough to shift it. Nothing touches it by hand after recording starts. It is never repositioned mid-task. A base that moves after recording starts ends the episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera is centered on the rug. Its frame includes the whole rug from binding to binding, the bare platform round it, the bin with its open top, and the dock with the vacuum in it.
3. The camera reads the rug from the front at an angle where the crumbs on it are visible, and a cleaned strip can be told from a strip still carrying crumbs. If a glare patch hides part of the rug, move the camera until the whole rug reads.
4. The camera reads the binding round the rug. You can see whether the edge pass ran tight to it.
5. The camera reads the nozzle from the front. You can see whether it is flat on the rug.
6. The camera reads the power light on top of the vacuum, so a running vacuum can be told from a stopped one.
7. The camera reads the underside of the dust cup while the vacuum hangs over the bin, so an open flap can be told from a shut one.
8. Both arms are at home with grippers open.
9. The platform, the dock, and the bin are fixed firm. They do not move, rock, or slide when the nozzle is pushed on the rug or the vacuum is pressed into the dock.
10. The **right arm** reaches the dock, the whole rug from the front edge to the back edge and from the left binding to the right binding, the switch spot, and the space over the bin. It reaches all of them without extending to a joint limit and without the wrist turning past a quarter turn.
11. The **left arm** reaches the power button at the switch spot and both sides of the dust cup while it hangs over the bin. It reaches both without extending to a joint limit. It never needs to reach the dock or any part of the rug beyond the switch spot.
12. Each arm comes at the platform **from the front, level**: in over the front edge and back out the same way. No part of the arm passes over the dock or the bin on its way to the rug.

13. Both arms work at the switch spot and over the bin at the same time without touching each other, with the left gripper left of the right gripper.
14. If a place cannot be reached, re-park the base by the Base positioning steps until lines 10 to 13 hold.

Materials checklist

1. The **platform** is one flat, raised floor section standing in front of the base, fixed so it cannot slide. Its top is level and about as high as a work table, so the arms reach it without reaching down to the floor. It is the only work surface in frame.
2. The platform is wide enough to hold the bin at its left, the rug in its middle-right, and the dock at its right end, with a clear gap between each of them.
3. The **rug** is a small, flat, short-pile rug with a **bound edge** all round. It lies flat on the platform with no fold, curl, or wrinkle, and its four edges run square to the front edge of the platform.
4. The rug has a non-slip backing. It does not slide, bunch, or lift when the nozzle is pushed across it.
5. The rug is narrow enough that **three lanes of the nozzle with overlap cover it from the left binding to the right binding**. It is shallow enough that the right gripper runs the nozzle from the front edge to the back edge in one move without extending.
6. The rug carries a **light, even scatter of dry crumbs** over its whole face, from binding to binding. The crumbs read on camera from the front. There is nothing on the rug a nozzle cannot lift: no thread, no wrapper, no wet patch, no object.
7. The rug is plain and one colour, so a cleaned strip reads against a strip still carrying crumbs.
8. The **handheld vacuum** is a cordless handheld with a **handle** at the back, a **dust cup** in the middle, and a fixed, wide, flat **nozzle** at the front. It is charged enough to run the whole pattern at full power.
9. The handle sits inside one gripper's opening, so the **right gripper** closes on it and the vacuum hangs level without sagging or turning in the grip. The nozzle sits flat on the rug when the handle is held level. One gripper carries the vacuum on its own without help.
10. The **power button** is a latching push button on top of the body, just in front of the handle. One press starts the vacuum and one press stops it. It does not need to be held. A **power light** beside it shows when the vacuum is running.
11. The **dust cup** is clear, so crumbs inside it are visible. Its **cup flap** is a hinged door on the underside of the cup. It is held shut by a **release catch** on the **left side** of the cup. One press on the catch lets the flap swing open downward. Pushing the flap up against the cup shuts it with a click.
12. The dust cup is empty at episode start. The flap is shut.
13. The **dock** is a fixed cradle at the right end of the platform with a spring **clip**. The vacuum sits in it **level**, nozzle pointing away from the base and handle nearest the base. It lifts

straight up out of the clip and lowers straight back in. Pressing it down seats it with a **click**, and a seated vacuum does not lift or rock when the gripper opens.

14. The vacuum sits clicked into the dock at episode start, power light off.
15. The **bin** is an open-top bin standing on the platform at the left, in front of the left arm, its right side about level with the middle of the base. Its open top is wider than the dust cup, so the crumbs fall inside it and not on its rim. It is empty at episode start.
16. The bin is heavy enough, or fixed, so it does not slide or tip when the flap is pushed shut above it.
17. The floor under the base is level and clear.

Workspace layout

Nothing anywhere is marked or taped out. You judge every place below by eye against the platform, the rug, and the tools already in place.

- **Platform:** the raised floor section in front of the base. Everything the episode touches stands on it. It is never leaned on and never pushed.
- **Rug:** the rug being vacuumed, lying flat in the middle-right of the platform, in front of the right arm. Its **left binding** lines up with the middle of the base.
- **Binding:** the bound edge that runs round the rug. The four **edges** of the rug are the front edge (nearest the base), the back edge, the left edge, and the right edge.
- **Corners:** the four places where two edges of the rug meet: front-left, back-left, back-right, front-right.
- **Switch spot:** the front-left corner of the rug. The nozzle rests here, flat, while the power button is pressed. It is the only place the left gripper reaches on the rug.
- **Bin:** the open-top bin on the platform to the left of the rug, in front of the left arm, with a clear gap to the rug. The dust cup is opened only over its open top.
- **Dock:** the cradle at the right end of the platform, to the right of the rug, with a clear gap to the rug. The vacuum starts and ends the episode clicked into it. **Right gripper only.**

Arm lanes

The two arms never cross. This holds for the whole episode.

- The **right gripper works the right and the middle:** the dock, the whole rug, the switch spot, and the space over the bin while it holds the vacuum there. The **left gripper works the left:** the bin and the two places on the vacuum it presses, the power button and the dust cup.
- The **left gripper always stays left of the right gripper.** Neither arm reaches over, under, around, or past the other. Neither arm reaches across the front of the other arm's body.
- The **left arm never goes onto the rug** past the switch spot and never goes near the dock.

- The **right gripper stays in front of the base** while it works the rug. It never swings out past the dock or over the bin on its way to the rug.
- Nothing is handed from gripper to gripper. Only the right gripper ever holds the vacuum.

Arm assignments

- **Right gripper. The tool gripper.** Lifts the vacuum out of the dock. Rests the nozzle at the switch spot for the switch-on. Runs all three lanes and the edge pass. Carries the vacuum back to the switch spot for the switch-off. Holds it level over the bin for the emptying. Lowers it into the dock and presses it until it clicks. It never presses the power button and never touches the cup flap or the release catch.
- **Left gripper. The button gripper.** Presses the power button once at the start and once at the end. Presses the release catch over the bin. Pushes the cup flap shut. It never touches the handle, the nozzle, the rug, the dock, or the bin.

Vocabulary

- **In situ:** the rug is worked where it lies. Nothing is taken away from the platform, and nothing about the platform is rearranged for the task.
- **Passive base:** the base has no drive. It is pushed by hand to its park before recording, locked there, and it does not move at all during the episode.
- **Front approach:** the gripper comes in over the front edge of the platform, level, and goes back out the same way. It never swings down from above and never passes over the dock or the bin on its way to the rug. Every action on the platform is a front approach.
- **Handle:** the grip at the back of the vacuum where the right gripper closes. A gripper closes on the handle only. It never closes on the body, the cup, or the nozzle.
- **Nozzle:** the wide, flat mouth at the front of the vacuum that touches the rug. **Nozzle flat** means the whole width of the mouth touches the rug at once. No end is lifted and no side rides up.
- **Power button:** the latching button on top of the body, just in front of the handle. One press starts the vacuum, one press stops it. **Power light** is the light beside it that shows while the vacuum runs. **Unvalidated:** button position and force.
- **Running:** the power light is on and the vacuum is sucking. **Stopped:** the power light is off.
- **Dust cup:** the clear middle section of the vacuum where the crumbs collect.
- **Cup flap:** the hinged door on the underside of the dust cup. **Release catch** is the button on the left side of the cup that lets the flap swing open. **Flap shut** means the flap sits against the cup and has clicked. **Unvalidated:** flap and catch design.
- **Dock:** the cradle at the right end of the platform. **Clip** is the spring in it that holds the vacuum. **Clicked in** means the vacuum has been pressed down until the clip clicks, sits

level in the dock, and does not lift or rock when the gripper opens. **Unvalidated:** lift-out and press-in motion.

- **Switch spot:** the front-left corner of the rug, where the nozzle rests for every press of the power button.
- **Lane:** one run of the nozzle along the rug from the front edge straight back to the back edge, or from the back edge straight out to the front edge. The nozzle stays flat the whole way, in one steady push.
- **Steady push:** the nozzle moves at one speed. It does not stop, jerk, speed up, lift, or turn.
- **Step over:** at the end of a lane, the nozzle is slid sideways to the start of the next lane without leaving the rug. It is a slide, not a lift.
- **Overlap:** the new lane is set so that **about a third of the nozzle sits on the cleaned strip** the last lane left. The rest sits on rug still carrying crumbs. You judge it by eye against the line between clean and crumbs.
- **Cleaned strip:** the band of rug a finished lane has left behind. It reads free of crumbs from the front, against the crumbs beside it.
- **Missed band:** a stretch of rug between two cleaned strips, or at an edge, that still carries crumbs. It is what the lane check looks for.
- **Clear:** no crumb is visible on that stretch of rug when it is read from the front. This is what a finished lane and a finished edge both look like.
- **Edge pass:** one unbroken run of the nozzle round all four edges of the rug, with the outer side of the nozzle kept tight to the binding, and the nozzle never turned.
- **Tight to the binding:** the outer end or side of the nozzle runs along the binding, with no missed band left between the nozzle and the binding, and without the nozzle going off the rug onto the bare platform.
- **Slide sideways:** the nozzle is moved along its own width, left or right, while it stays flat and pointing the same way. This is how the front edge and the back edge are run.
- **Level and low:** the vacuum is carried with the body level and the nozzle just above the platform, never high, and never tipped.
- **Loaded:** pressed, leaned on, or pushed hard enough to move. The platform, the rug, the dock, and the bin are never loaded.

Steps

Step 1: Unclip the vacuum from the dock

Goal: the **right gripper** holds the vacuum level by its handle, clear of the dock.

- With the **right gripper**, come in from the front and close on the **handle** of the docked vacuum. Do not touch the body, the cup, or the nozzle.

- Lift the vacuum **straight up** out of the clip until the whole vacuum is clear of the dock. If the clip holds, lower it back and lift straight up again. Do not lever it sideways and do not twist it out.
- Hold the vacuum **level**, nozzle pointing away from the base, and carry it **level and low** along the front of the platform to the rug. Do not carry it over the bin. Do not swing it over the dock.

Check: the vacuum hangs level in the **right gripper**, clear of the dock, power light off. The dock has not moved. If the vacuum hangs tipped or turns in the grip, set it back in the dock with the **right gripper**, close on the handle again, and lift again.

Expected state: vacuum in the right gripper, stopped. Dock empty. Rug untouched, crumbs over the whole of it. Bin empty. Left arm at home.

Step 2: Switch the vacuum on

Goal: the vacuum is **running**, with the nozzle flat at the switch spot.

This is a step where both grippers work at once. The **right gripper holds the vacuum still**. The **left gripper presses the button**.

- With the **right gripper**, lower the nozzle straight down onto the rug at the **switch spot**, the front-left corner. Set it **nozzle flat**, with its left end just inside the left binding and its front side just inside the front binding. Hold the handle still there.
- With the **left gripper**, come in from the front, level, and press the **power button** straight down **once**. Do not touch the handle, the right gripper, or the cup.
- Lift the **left gripper** straight up, clear of the vacuum, and draw the arm out to the front. The left arm stays at home until Step 5.
- With the **right gripper**, keep the nozzle still at the switch spot until the left arm is clear.

Check: the power light is on and the nozzle is flat at the switch spot. If the light is off, press the button once more with the **left gripper**. If the nozzle moved during the press, set it again at the switch spot with the **right gripper** before lane 1.

Expected state: vacuum running, nozzle flat at the front-left corner of the rug, right gripper on the handle. Left arm at home. Crumbs still over the whole rug.

Step 3: Vacuum the rug in three lanes

Goal: the whole rug is vacuumed in three overlapping lanes, left to right, run as one unbroken pattern.

Run 3.1 and 3.2 in turn: **lane 1, step over, lane 2, step over, lane 3**. There is no step over after lane 3.

The nozzle does not leave the rug at any point in Step 3. The left arm stays at home, clear of the rug, for the whole of it. The right gripper stays in front of the base and does not swing out toward the dock.

3.1 Run one lane

- With the **right gripper**, push the nozzle **straight back from the front edge to the back edge** in one **steady push**, keeping the nozzle flat on the rug the whole way. This is how **lane 1** and **lane 3** run.
- On **lane 2**, pull the nozzle **straight out from the back edge to the front edge** in the same way.
- Stop the nozzle **just inside the back binding** on the way back, and **just inside the front binding** on the way out. Do not run the nozzle off the rug onto the bare platform.
- Do not lift the nozzle. Do not stop mid-lane. Do not jerk it. Do not turn it as it runs.
- Do not extend the arm to reach the back edge. If the back edge is out of reach, the park is wrong: end the episode and re-park.

Check: the strip the nozzle has just laid reads **clear** from the front, with no missed band left in it. If crumbs still show in the strip, run that same lane again with the **right gripper**, once.

Expected state: one more cleaned strip on the rug. Nozzle flat on the rug at the end of that lane, vacuum running. Left arm at home.

3.2 Step over to the next lane

- With the **right gripper**, slide the nozzle sideways **to the right**, keeping it flat on the rug. Do not lift it.
- Set it so about a **third of the nozzle sits on the cleaned strip** the last lane left and the rest sits on crumbs.
- On the step over into **lane 3**, also set the nozzle so its **right end is just inside the right binding**, not off the rug.

Check: the nozzle sits flat, with about a third of it on the cleaned strip and the rest on crumbs. If it is set clear of the cleaned strip, or so far onto it that almost no crumbs are under it, slide it again with the **right gripper**.

Expected state after the third lane: the whole rug reads clear from the left binding to the right binding and from the front edge to the back edge, apart from a thin band of crumbs along the bindings. The nozzle is flat on the rug at the back-right corner, and the vacuum is running.

Check after the third lane: read the whole rug from the front. If a missed band shows between any two lanes, slide the nozzle onto it with the **right gripper** and run one more lane over that band, once, by Steps 3.1 and 3.2.

Step 4: Run the edge pass

Goal: all four edges of the rug are vacuumed tight to the binding, in one unbroken run, with the vacuum still running.

The edge pass starts where lane 3 ends, at the **back-right corner**, and goes round the rug to the left. The nozzle **is never turned**. It keeps pointing away from the base for the whole pass. Along the back edge and the front edge it is **slid sideways**. Along the left edge and the right edge it is pushed or pulled straight.

- With the **right gripper**, run the nozzle round the rug in one unbroken move, keeping it flat and its outer side or end **tight to the binding** it is on, in this order:
 - along the **back edge**, slid sideways from the back-right corner to the back-left corner, with the back side of the nozzle tight to the back binding;
 - down the **left edge**, pulled straight out from the back-left corner to the front-left corner, with the left end of the nozzle tight to the left binding;
 - along the **front edge**, slid sideways from the front-left corner to the front-right corner, with the front side of the nozzle tight to the front binding;
 - up the **right edge**, pushed straight back from the front-right corner to the back-right corner, with the right end of the nozzle tight to the right binding.
- At each corner, **change direction without turning the nozzle**. Do not twist the wrist. Do not lift the nozzle off the rug to reset it.
- Keep the nozzle **on the rug**. No part of it goes off the binding onto the bare platform.
- Do not lean on the platform. Do not push hard enough to bunch or move the rug.
- Keep the right arm in front of the base for the whole pass. Do not swing it out toward the dock.

Check: all four edges read **clear** from the front, with no missed band left between the cleaned rug and the binding, and the rug still lies flat and square. If a missed band shows along an edge, run the nozzle along that edge again with the **right gripper**, once, in the same way.

Expected state: the whole rug reads clear from binding to binding and corner to corner. The nozzle is flat on the rug at the back-right corner. The vacuum is still running. The bin and the dock are where they started.

Step 5: Switch the vacuum off

Goal: the vacuum is **stopped**, with the nozzle flat at the switch spot.

This is the one time the nozzle leaves the rug while the vacuum runs. It is lifted, carried, and set down. It is not slid back across the rug.

- With the **right gripper**, lift the nozzle **straight up** off the rug at the back-right corner, just clear of the pile.

- Carry the vacuum **level and low** over the rug to the **switch spot** and lower the nozzle straight down onto the rug there, **nozzle flat**. Hold the handle still.
- With the **left gripper**, come in from the front, level, and press the **power button** straight down **once**. Do not touch the handle, the right gripper, or the cup.
- Lift the **left gripper** straight up, clear of the vacuum, and draw the arm out to the front.
- With the **right gripper**, keep the nozzle still at the switch spot until the left arm is clear.

Check: the power light is off and the nozzle is flat at the switch spot. If the light is still on, press the button once more with the **left gripper**. Do not lift the vacuum off the rug while the light is on.

Expected state: vacuum stopped, nozzle flat at the front-left corner, right gripper on the handle. Rug clear from binding to binding. Left arm clear. Dust cup holding the crumbs.

Step 6: Empty the dust cup into the bin

Goal: the crumbs are in the bin, the dust cup is empty, and the cup flap is shut.

This is a step where both grippers work at once. The **right gripper holds the vacuum still over the bin**. The **left gripper works the flap**. The vacuum stays **stopped** for the whole step.

6.1 Open the flap over the bin

- With the **right gripper**, lift the vacuum **straight up** off the switch spot and carry it **level and low** to the left, until the **dust cup** is above the middle of the bin's open top, just above the rim. Keep the vacuum level, underside down, with the **release catch facing the left arm**. Hold it still there.
- With the **left gripper**, come in from the left, level, and press the **release catch** once. The flap swings open downward and the crumbs fall into the bin.
- With the **right gripper**, hold the vacuum still over the bin for **2 seconds** so the last crumbs fall inside. Do not shake it, tap it, or tip it.

Check: the flap hangs open, the dust cup reads empty, and the crumbs are in the bin. If crumbs are still in the cup after 2 seconds, hold it still for 2 seconds more with the **right gripper**. If crumbs have fallen on the platform or the bin rim, leave them; they are cleared at the reset.

6.2 Shut the flap

- With the **right gripper**, lift the vacuum **straight up** until the open flap hangs clear above the bin rim. Keep the vacuum level and still.
- With the **left gripper**, come in from below the flap, level, and push the flap **up against the cup** until it clicks. Push only the flap. Do not push the cup, the body, or the right gripper.
- Lift the **left gripper** clear and draw the arm out to the front.

Check: the flap sits shut against the cup and does not hang open when the left gripper is clear. If it hangs open, push it up again with the **left gripper** until it clicks.

Expected state: dust cup empty, flap shut, crumbs in the bin. Vacuum level in the right gripper above the bin, stopped. Left arm clear. Rug clear.

Step 7: Dock the vacuum and click it in

Goal: the vacuum is **clicked in** to the dock, level, with the power light off.

- With the **right gripper**, carry the vacuum **level and low** along the front of the platform to the right, past the rug, until it is above the dock. Do not carry it over the rug and do not swing it high.
- Line it up over the clip, nozzle pointing away from the base and handle nearest the base, as it sat at the start.
- Lower it **straight down** into the dock until it stops on the clip. Then press straight down once until the clip **clicks**. Do not lower it at an angle and do not force it.
- If it does not click, lift it straight up, line it up again, and lower it again. Never push it in sideways.
- Open the **right gripper** and draw the arm out to the front. Do not let go before the click.

Check: the vacuum sits level in the dock, clicked in, and does not lift, rock, or slide when the **right gripper** opens. The power light is off. If it rocks or sits proud of the clip, close on the handle again with the **right gripper**, lift it straight up, and dock it again.

Expected state: vacuum clicked into the dock, stopped. Rug clear. Crumbs in the bin. Both grippers empty.

Step 8: End the episode

- Confirm the rug. It reads clear from the left binding to the right binding and from the front edge to the back edge, with no missed band and no crumb left on it. It lies flat and square as it did at the start.
- Confirm the edges. All four read clear, with no missed band left along the binding.
- Confirm the bin. The crumbs are inside it and none lie on its rim.
- Confirm the vacuum. It sits clicked into the dock, level, flap shut, power light off.
- Confirm the platform. It stands as it did at the start and nothing on it has moved.
- Confirm the base has not moved.
- Correct any failed check before ending.
- Return both arms home with grippers open. Then stop recording.

After the episode: reset the workspace

This reset is not recorded. The base stays parked and locked through the reset, and is only pushed away by hand once the station is set for the next episode.

1. Empty the bin and stand it back on the platform at the left, in front of the left arm, with its right side about level with the middle of the base and a clear gap to the rug.
2. Lift the vacuum out of the dock, open the cup flap, and check the dust cup is empty. Wipe out any crumbs stuck inside. Shut the flap until it clicks.
3. Check the charge. If the vacuum will not run the whole pattern at full power, charge it or swap in a charged one of the same model.
4. Dock the vacuum level, nozzle pointing away from the base, handle nearest the base, and press it until it clicks. Check the power light is off.
5. Brush any crumbs off the bare platform and the bin rim.
6. Straighten the rug so it lies flat and square with its left binding at the middle of the base. Then put a fresh light, even scatter of dry crumbs over its whole face, from binding to binding. Check it reads on camera from the front.
7. Replace a rug that curls, wrinkles, sheds, or slides under the nozzle.
8. Replace a vacuum whose handle turns in the grip, whose nozzle no longer sits flat when the handle is level, whose power button no longer latches, or whose flap no longer clicks shut.
9. Replace a dock whose clip no longer clicks or no longer holds the vacuum level and still.
10. Replace a bin that slides or tips when the flap is pushed shut above it.
11. Check the platform, the dock, and the bin are still fixed firm and do not move when pushed.
12. Check the base is still locked and parked square. Then run the Base positioning steps and both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Base moved during the episode

- **Visible cue:** the platform shifts in frame, the front edge changes angle or size in frame, or the base rolls, creeps, or turns at any point after recording starts.

- **SOP rule broken:** Steps 1 to 8. The base is parked and locked before recording and stays still for the whole episode.
- **Coaching note:** park it, lock it, push-test it, then start recording. A base that drifts is a lost episode.

Violation: Platform, rug, dock, or bin loaded

- **Visible cue:** the platform, the dock, or the bin moves, rocks, or slides. A gripper, wrist, or forearm leans or rests on one of them. The rug bunches, slides, or lifts under the nozzle. Or the vacuum is pressed into the dock hard enough to move it.
- **SOP rule broken:** Steps 1 to 7. The platform, the rug, the dock, and the bin are never loaded, and the nozzle is pushed only hard enough to keep it flat.
- **Coaching note:** the furniture holds nothing up for you. Light pressure, and let the arm carry its own weight.

Violation: Not a front approach

- **Visible cue:** an arm swings down onto the platform from above instead of coming in level over the front edge. Or the vacuum is carried over the bin or over the dock on its way to or from the rug, or carried high instead of level and low.
- **SOP rule broken:** Steps 1 to 7. Every action on the platform is a front approach, and the vacuum travels level and low along the front of the platform.
- **Coaching note:** in over the front, do the work, out over the front. Nothing goes in from above.

Violation: Wrong order

- **Visible cue:** the vacuum goes onto the rug before it is unclipped. A lane is run before the power light is on. The edge pass is run before the third lane. The vacuum goes toward the bin while the light is on. The flap is opened before the switch-off. Or the vacuum is docked before the cup is emptied.
- **SOP rule broken:** Steps 1 to 7. Unclip, switch on, three lanes, edge pass, switch off, empty, dock.
- **Coaching note:** unclip, on, lanes, edges, off, empty, dock. Nothing runs early and nothing is skipped.

Violation: Vacuum unclipped wrongly

- **Visible cue:** the **right gripper** closes on the body, the cup, or the nozzle instead of the handle. The vacuum is levered, twisted, or pulled sideways out of the clip. It is lifted at an angle. Or it hangs tipped or turns in the grip after the lift.
- **SOP rule broken:** Step 1. Close on the handle only and lift the vacuum straight up out of the clip.

- **Coaching note:** handle only, straight up. If the clip holds, go down and come straight up again.

Violation: Power button pressed wrongly

- **Visible cue:** the **left gripper** presses the button while the nozzle is off the rug or away from the switch spot. It presses more than once when the light has already changed. It touches the handle, the right gripper, or the cup. The **right gripper** presses the button. Or the vacuum moves under the press.
- **SOP rule broken:** Steps 2 and 5. The right gripper holds the nozzle flat and still at the switch spot, and the left gripper presses the power button straight down once.
- **Coaching note:** nozzle down at the corner, hold it, then one press. The button is the left gripper's only job on the rug.

Violation: Vacuum on the rug while stopped, or off the rug while running

- **Visible cue:** the nozzle is pushed along the rug with the power light off. Or the vacuum is lifted off the rug and carried toward the bin, or anywhere else, with the power light on. The one lift allowed is the carry from the back-right corner to the switch spot in Step 5.
- **SOP rule broken:** Steps 2 to 5. The vacuum is running for every lane and the edge pass, and it is switched off at the switch spot before it leaves the rug.
- **Coaching note:** on at the corner, work, back to the corner, off. A running vacuum in the air is going the wrong way.

Violation: First lane not set right

- **Visible cue:** the nozzle starts somewhere other than the front-left corner. It starts with an end off the rug, or with a gap to the left binding. It starts with an end lifted. Or it starts pointing across the rug instead of away from the base.
- **SOP rule broken:** Step 2. Set the nozzle flat at the switch spot, its left end just inside the left binding and its front side just inside the front binding.
- **Coaching note:** set it, look at it, then push. A pattern that starts wrong cannot finish right.

Violation: Lane run wrong

- **Visible cue:** the nozzle stops mid-lane, jerks or changes speed, turns as it runs, is dragged sideways during the push, or an end of the nozzle rides up off the rug. Or the arm extends to reach the back edge.
- **SOP rule broken:** Step 3.1. Run each lane in one steady push with the nozzle flat the whole way.
- **Coaching note:** one steady push, all the way. Stopping mid-lane leaves a band you cannot see until you look for it.

Violation: Nozzle lifted off the rug during the pattern

- **Visible cue:** the nozzle comes off the rug at any point between the start of lane 1 and the end of the edge pass. It is lifted to reset a lane, lifted at a step over, lifted at a corner, or hopped across the rug.
- **SOP rule broken:** Steps 3.1 to 4. The nozzle stays on the rug from the start of lane 1 to the end of the edge pass. It is lifted only once, in Step 5.
- **Coaching note:** slide, never lift. A nozzle that hops leaves a gap.

Violation: Lanes not overlapped

- **Visible cue:** a missed band is left between two cleaned strips. A lane is set clear of the strip before it. Or a lane is set so far onto the cleaned strip that it covers almost no crumbs.
- **SOP rule broken:** Step 3.2. Set each new lane so about a third of the nozzle sits on the cleaned strip the last lane left.
- **Coaching note:** a third on the clean, the rest on the crumbs. Read the clean line before you set the nozzle.

Violation: Lanes worked in the wrong direction, or a lane left out

- **Visible cue:** the step overs run right to left. The lanes run out of order. Fewer or more than three lanes are run. Or the rug at a binding is never covered by a lane.
- **SOP rule broken:** Steps 2 to 3.2. Run three lanes from the left edge of the rug rightward: lane 1, step over, lane 2, step over, lane 3.
- **Coaching note:** left side first, work right, finish on the right. Three lanes, two step overs, one pattern.

Violation: Nozzle run off the rug

- **Visible cue:** any part of the nozzle goes past the binding onto the bare platform during a lane, a step over, or the edge pass.
- **SOP rule broken:** Steps 3.1, 3.2, and 4. Stop the nozzle just inside the binding on every lane, and keep it on the rug for the whole edge pass.
- **Coaching note:** the rug is the whole run. The bare platform is not part of it.

Violation: Edge pass broken up or run out of order

- **Visible cue:** the nozzle leaves the rug between edges. The edge pass is run as separate strokes instead of one unbroken run. The edges are run in an order other than back, left, front, right. Or an edge is skipped.
- **SOP rule broken:** Step 4. Run the nozzle round the rug in one unbroken move: back edge, left edge, front edge, right edge.
- **Coaching note:** one lap round the outside, in one go. Lifting between edges is how an edge gets missed.

Violation: Edge not run tight to the binding

- **Visible cue:** a missed band is left between the cleaned rug and the binding. The nozzle runs along an edge with a gap to the binding. Or a corner of the rug is cut instead of run into.
- **SOP rule broken:** Step 4. Keep the outer side or end of the nozzle tight to the binding it is on, and run into each corner.
- **Coaching note:** the nozzle follows the binding, not near it. The corners are part of the run, not shortcuts.

Violation: Nozzle turned at a corner

- **Visible cue:** the **right gripper** twists at the wrist to turn the nozzle at a corner, so the nozzle points across the rug instead of away from the base. Or the wrist reaches a limit and the nozzle stalls or skids.
- **SOP rule broken:** Step 4. Change direction at each corner without turning the nozzle: slide it sideways along the back and front edges, push or pull it straight along the left and right edges.
- **Coaching note:** the nozzle always points the same way. Slide it along the short edges, push it along the long ones.

Violation: Switch-off carry done wrongly

- **Visible cue:** after the edge pass the nozzle is slid back across the rug to the switch spot instead of lifted and carried. It is carried high, or tipped. It is set down somewhere other than the switch spot. Or it is set down with an end off the rug or lifted.
- **SOP rule broken:** Step 5. Lift the nozzle straight up at the back-right corner, carry the vacuum level and low to the switch spot, and set the nozzle down flat there.
- **Coaching note:** up, across, down at the corner. The rug is clean now, so do not slide over it.

Violation: Dust cup emptied wrongly

- **Visible cue:** the release catch is pressed while the cup is not above the open top of the bin. The **right gripper** presses the catch. The vacuum is tipped, shaken, or tapped over the bin. The vacuum is not held still for 2 seconds. The catch is pressed while the power light is on. Or crumbs are let fall anywhere but into the bin.
- **SOP rule broken:** Step 6.1. Hold the stopped vacuum level and still with the cup above the middle of the bin, press the catch once with the left gripper, and hold still for 2 seconds.
- **Coaching note:** cup over the bin, one press, hold still, count two. The crumbs fall on their own.

Violation: Flap not shut, or shut wrongly

- **Visible cue:** the flap is pushed shut while it still hangs inside the bin rim. The **left gripper** pushes the cup, the body, or the right gripper instead of the flap. The **right gripper** shuts

the flap by pressing the vacuum down onto the bin. The flap is left hanging open when the left arm draws out. Or the vacuum is carried toward the dock with the flap open.

- **SOP rule broken:** Step 6.2. Lift the vacuum until the flap is clear of the rim, then push the flap up against the cup with the left gripper until it clicks.
- **Coaching note:** lift it clear, then push the flap up until you hear the click. Nothing else gets pushed.

Violation: Vacuum docked wrongly

- **Visible cue:** the vacuum is lowered into the dock at an angle, pushed in sideways, or forced. It is set in the dock pointing the wrong way. The **right gripper** opens before the click. The vacuum sits proud of the clip, rocks, lifts, or slides after the gripper opens. Or it is carried to the dock over the rug or swung high.
- **SOP rule broken:** Step 7. Carry the vacuum level and low along the front of the platform, line it up over the clip, lower it straight down, press once until it clicks, and let go only after the click.
- **Coaching note:** line it up, straight down, press, click, then let go. Watch it sit before the arm leaves.

Violation: Required check not followed

- **Visible cue:** a check named in a step is skipped. Or a check is made and the fault it finds is left uncorrected: a power light that did not change, a missed band between lanes, a missed band along an edge, crumbs still in the cup, a flap hanging open, or a vacuum that rocks in the dock.
- **SOP rule broken:** Steps 1, 2, 3.1, 3.2, 4, 5, 6.1, 6.2, and 7. Run each check and correct what it finds by the retry written in that step.
- **Coaching note:** a check is not done until what it found has been put right.

Violation: Dropped, knocked over, or spilled

- **Visible cue:** the vacuum is dropped. The bin is knocked off its place or tipped. The dock is knocked. Crumbs are knocked off the rug onto the platform by an arm. Or an arm sweeps the bin or the dock on its way in or out.
- **SOP rule broken:** Steps 1 to 7. Nothing is dropped, knocked out of its place, or spilled, and every arm comes in and goes out clear of the bin and the dock.
- **Coaching note:** check the path and the landing place before the arm moves, and let go only when the thing is seated.

Violation: Wrong arm used, or arms crossed

- **Visible cue:** the **left gripper** touches the handle, the nozzle, the dock, the bin, or any part of the rug beyond the switch spot. The **right gripper** presses the power button, the release catch, or the flap. The left gripper goes to the right of the right gripper. One arm reaches

over, under, around, or past the other. Or the vacuum is handed from one gripper to the other.

- **SOP rule broken:** Steps 1 to 7. The right gripper holds the vacuum for the whole episode. The left gripper presses only the power button and works only the flap and catch. The left gripper always stays left of the right gripper.
- **Coaching note:** right gripper holds, left gripper presses. Each arm stays in its own lane.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a missed band or a crumb on the rug, an edge not run tight, crumbs on the platform or the bin rim, the vacuum out of the dock, not clicked in, or with its light on, the flap open, an arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 8. Confirm the rug, the edges, the bin, the vacuum, the platform, and the base. Then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Camera cannot read the rug:** glare on the rug, or a shadow from the arms, hides part of it, so whether it is clear cannot be judged.
- **Camera cannot read the binding,** so whether the edge pass ran tight cannot be judged.
- **Camera cannot read the power light,** so whether the vacuum was running cannot be judged.
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Base fault:** a brake or lock that will not hold, or a base that rolls, creeps, or turns with the lock set.
- **Platform not fixed:** a platform, dock, or bin that slides or rocks under a correct push.
- **Rug too large for one park:** a rug a correctly parked base cannot cover, so the back edge or the right binding cannot be reached without extending the arm.
- **Defective vacuum:** one that will not start or stop on one press, that loses power before the edge pass is done, whose nozzle will not sit flat when the handle is level, whose handle turns in the grip whatever the hold, or whose flap will not click shut.
- **Defective dock:** a clip that will not click, that will not hold the vacuum level, or that lets it rock after a correct press.
- **Defective rug:** one that curls, wrinkles, sheds, or slides on the platform under a correct push.

- **Defective bin:** one that tips or slides when the flap is pushed shut above it, or whose open top is too narrow for the crumbs to fall inside.
- **Debris that will not lift:** a thread, a wrapper, a wet patch, or a crumb trodden into the pile that a correct lane and a correct edge pass will not clear.
- **A place turns out to sit outside its arm's comfortable reach** with the base correctly parked, so the dock, the switch spot, the space over the bin, or a corner of the rug cannot be reached without extending or folding the arm.

Annotation subtasks (from SOP)

1. Lift the vacuum out of the dock
2. Rest the nozzle at the switch spot and press the power button on
3. Run one lane across the rug
4. Step the nozzle over to the next lane
5. Run the edge pass round the rug
6. Carry the vacuum back to the switch spot and press the power button off
7. Hold the vacuum over the bin and open the cup flap
8. Push the cup flap shut
9. Dock the vacuum and press it until it clicks
10. Return both arms home and end the episode